



Degree, Semester & Branch : VI Semester B.E. ECE A&B

Course Code & Title : CEC351 Rocketry and Space Mechanics

Name of the Faculty member: Mrs.G.Subhashini

Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT I ORBITAL MECHANICS				
1.	02.02.2024	Two body Problem	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/
		<i>Two body problem</i> In universe, each and every body is attracted to each other. The derivation in the solution of two body problem may occur, when the bodies are not stars. Universal Gravitation Law states, $F_{12} = -G \frac{m_1 m_2}{r^2}$ where, r is the distance between the two bodies. $\vec{r}$ = position vector, which represents the acceleration vector ($\vec{a}$).		
		<i>Two BODY MOTION</i> Every body is attracted to each other. Since the distance between the bodies is large the motion of two bodies is influenced by their gravitational forces alone when the other body effects are negligible. Consider two homogeneous spheres: $F_{21} = \frac{G m_1 m_2}{r^2}$ (by Newton's law of gravitation) $F_{21} = m_2 a_2$ $\therefore m_2 a_2 = \frac{G m_1 m_2}{r^2}$ [$F_{12} = -F_{21}$]		



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Department of Electronics and Communication Engineering

Academic Year: 2023 – 2024(Even Semester)

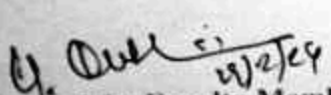
INNOVATIVE TEACHING METHODS

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UNIT II SATELLITE DYNAMICS				
1.	19.02.2024	Hohmann orbit	One Minute Speech	https://cft.vanderbilt.edu/wp-content/uploads/sites/59/Active-Learning.pdf
				


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Innovative Practice Description

- **Unit / Topic:** III / One dimensional and two dimensional rocket motions homogeneous gravitational fields
- **Course Outcome:** CO4
- **Activity Chosen:** Flipped Class room
- **Justification:**
 - The topic of One dimensional and two dimensional rocket motions homogeneous gravitational fields is chosen for the flipped classroom approach because it encompasses a comprehensive understanding of various rocket motions with respect to gravitational field. This topic covers key areas such as One Dimensional Rocket motion and two dimensional rocket motion with respect to gravitational field. These elements are crucial for staging of rocket and align well with self-directed learning through exploration and practical application.
- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO12
CO3	3	3	3	2	2	2	2

• PO / PSO mapped

	PSO1	PSO2
Innovative practice	3	2
Justification for correlation	The students will be able to analyze the rocket motion. Thus, PSO1 is mapped at level 3.	The understanding of rocket motion and trajectory calculations, the problem-solving and analytical skills. Thus, PSO2 is mapped at level 2.

• Images / Screenshot of the practice:

Innovative Teaching Method Execution

One dimensional and two dimensional rocket motions homogeneous gravitational fields



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

❖ *Benefit of the practice:*

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of Buck, boost, buck boos analysis and design.

❖ *Challenges faced in implementation:*

Some of the students felt difficult to present because of lack of communication. Few groups done their presentation beyond their time limit. Few students in the group felt difficult to answer all the questions.

References:

1. Suresh. B N. & Sivan. K, "Integrated Design for Space Transportation System", Springer India; 2015.
2. Sutton, GP, "Rocket Propulsion Elements", John Wiley & Sons Inc., New York, 8th Edition, 2010.
3. Van de Kamp, "Elements of Astromechanics", Pitman Publishing Co., Ltd., London, 1980.

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UNIT IV ROCKET AERODYNAMICS

1.	25.03.2024	Performance at various altitudes	Role Play	https://ablconnect.harvard.edu/role-play-research
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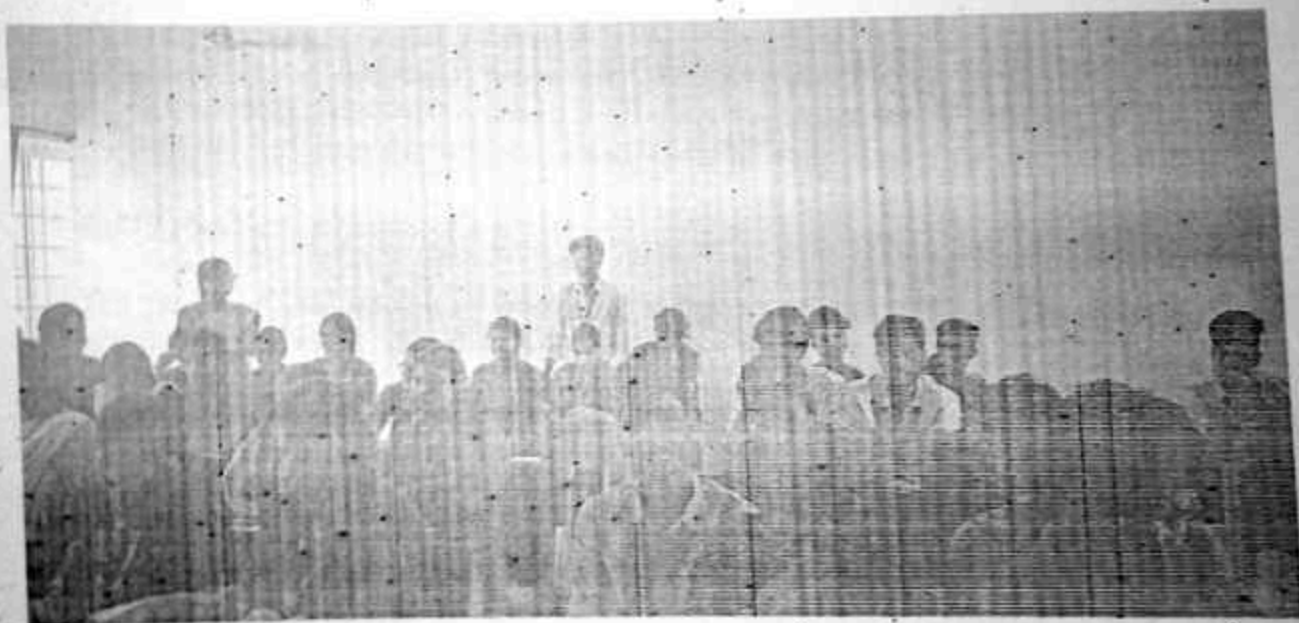
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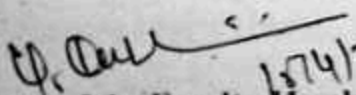
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UNIT V STAGING AND CONTROL OF ROCKET VEHICLES				
1.	15.04.2024	Multistage vehicle optimization	Head and talk	https://blog.hubspot.com/marketing/creative-exercises-better-than-brainstorming




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